Multiple Narrow-Band Signal Identification with MUSIC

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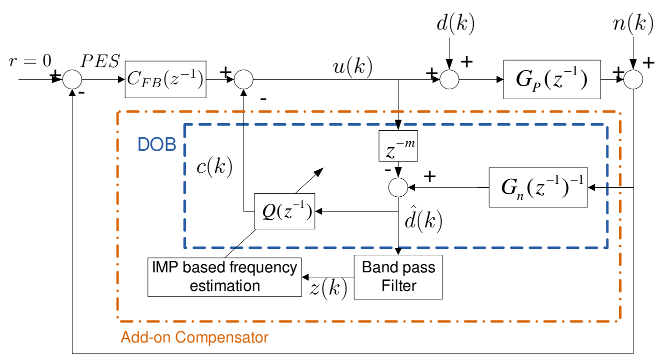
1 Motivation

1.1 Frequency Estimation-A Signal Processing Perspective

Frequency Estimation In the Presence of Noise

- Example Controller Structure



- Disturbance Observer: $\hat{d}(k)$ is a good estimate of $d(k)$; $z(k)$ is composed of narrow-band disturbances of interest.

- Goal: identify frequency of $\hat{d}(k)$ when it is mainly narrow-band signals.

1.2 MUSIC-A Statistical Signal Processing Point of View

MUSIC-Multiple Signal Classification. "Among the various high-resolution algorithms, MUSIC is the most promising" (M.I.T.'s Lincoln Lab 1989).

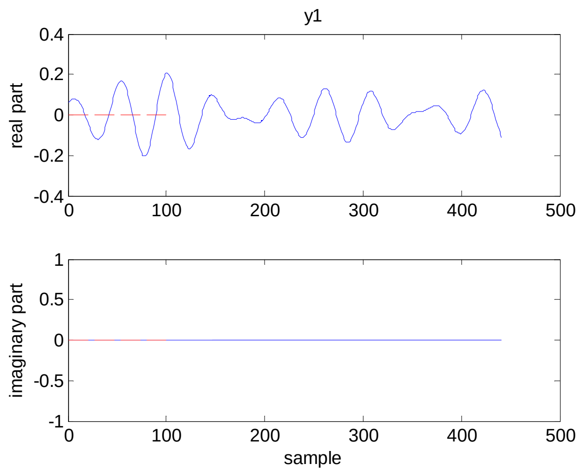
- Ref: Multiple emitter location and signal parameter estimation R Schmidt - IEEE Transactions on Antennas and Propagation, 1986 Cited by 3441

- Main idea: separate the narrow band signals from the additive (white) noise. (a subspace method)

2 MUSIC

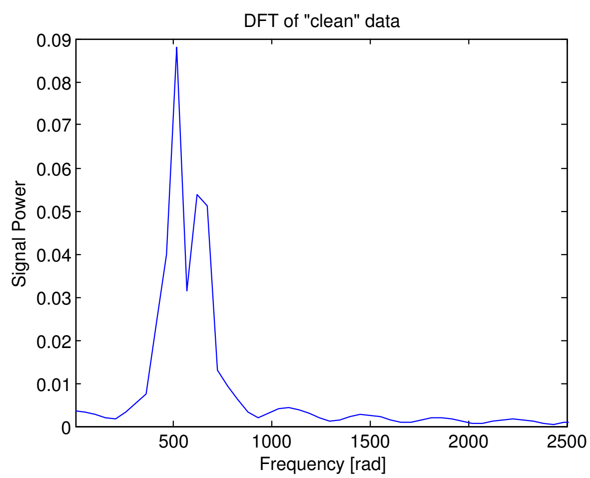
2.1 Why Not Simply apply DFT?

Narrow-band frequencies: 504 Hz and 648 Hz; signal length: 440 samples

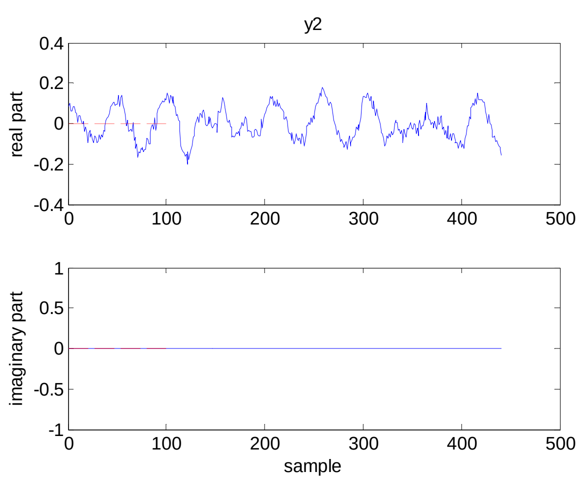


Relatively “clean” signal

Discrete time Fourier Transform (DFT):

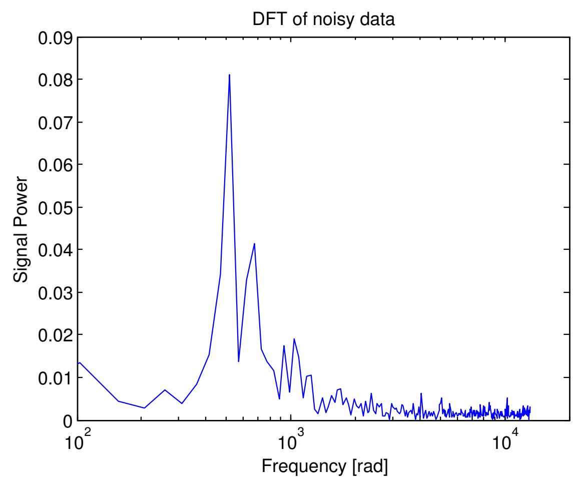


Now check the noisy version:



Signal in colored noise (actually the PES)

Discrete time Fourier Transform (DFT):



Peak frequencies from DFT of "clean" signal y1 (in Hz): 517.6 and 621.2

Peak frequencies from DFT of noisy signal y2 (in Hz): 517.6 and 672.9

True values (in Hz): 504 and 648

$\Longrightarrow$DFT won't give the correct frequencies when (1) the signal length is short; (2) the sinusoidals are not well separated in frequency.

A high resolution technique is required.

2.2 MUSIC: Problem Reformulation

Problem: multiple sinusoidals buried in white noise (matheamtically easy and rigorous to consider complex sinusoidals)

$$y\left[n\right] = \sum_{i=1}^{K} \alpha_i e^{j(\omega_i n + \phi_i)} + v\left[n\right] \tag{1}$$

where $\phi_i \in Unif\left[-\pi, \pi\right]$ and i.i.d. (Don't know the initial phase value), $v\left[n\right]$: i.i.d. Gaussian white noise.

Goal: estimate $\omega_i' s$

Matrix representation:

$$\begin{aligned} y\left[n\right] &= \sum_{i=1}^{K} \alpha_i e^{j(\omega_i n + \phi_i)} + v\left[n\right] \\ &= \left[1, 1, \ldots, 1\right] \times \begin{bmatrix} \alpha_1 e^{j(\omega_1 n + \phi_1)} \\ \alpha_2 e^{j(\omega_2 n + \phi_2)} \\ \vdots \\ \alpha_K e^{j(\omega_K n + \phi_K)} \end{bmatrix} + v\left[n\right] \end{aligned}$$

$$\begin{aligned} y\left[n-1\right] &= \sum_{i=1}^{K} \alpha_i e^{j(\omega_i (n-1) + \phi_i)} + v\left[n-1\right] \\ &= \left[e^{-j\omega_1}, e^{-j\omega_2}, \ldots, e^{-j\omega_K}\right] \times \begin{bmatrix} \alpha_1 e^{j(\omega_1 n + \phi_1)} \\ \alpha_2 e^{j(\omega_2 n + \phi_2)} \\ \vdots \\ \alpha_K e^{j(\omega_K n + \phi_K)} \end{bmatrix} + v\left[n-1\right] \end{aligned}$$

Introduce the matrix notation

$$\mathbf{y}[n] = \begin{bmatrix} y[n] \\ y[n-1] \\ \vdots \\ y[n-M+1] \end{bmatrix} \in \mathbb{C}^{M\times 1};\ \mathbf{v}[n] = \begin{bmatrix} v[n] \\ v[n-1] \\ \vdots \\ v[n-M+1] \end{bmatrix} \in \mathbb{C}^{M\times 1} \tag{2}$$

$$\mathbf{x}[n] = \begin{bmatrix} \alpha_1 e^{j(\omega_1 n+\phi_1)} & \alpha_2 e^{j(\omega_2 n+\phi_2)} & \ldots & \alpha_K e^{j(\omega_K n+\phi_K)} \end{bmatrix}^T \in \mathbb{C}^{K\times 1} \tag{3}$$

$$\begin{aligned} A &= \begin{bmatrix} 1 & 1 & \cdots & 1 \\ e^{-j\omega_1} & e^{-j\omega_2} & \cdots & e^{-j\omega_K} \\ \vdots & \vdots & \ddots & \vdots \\ e^{-j(M-1)\omega_1} & e^{-j(M-1)\omega_2} & \cdots & e^{-j(M-1)\omega_K} \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{a}(\omega_1) & \mathbf{a}(\omega_2) & \ldots & \mathbf{a}(\omega_K) \end{bmatrix} \end{aligned} \tag{4}$$

$$\mathbf{a}(\omega) = \begin{bmatrix} 1 \\ e^{-j\omega} \\ \vdots \\ e^{-j(M-1)\omega} \end{bmatrix} \tag{5}$$

We can then represent Eq. (1) as

$$\underbrace{\mathbf{y}[n]}_{M\times 1} = \underbrace{A}_{M\times K}\underbrace{\mathbf{x}[n]}_{K\times 1} + \underbrace{\mathbf{v}[n]}_{M\times 1} \tag{6}$$

Key step: forming the covariance matrices of Eq. (6)

$$R_y = E\{\mathbf{y}[n]\mathbf{y}[n]^*\} = AE\{\mathbf{x}[n]\mathbf{x}[n]^*\}A^* + \sigma^2 I_{M\times M} \tag{7}$$

where we have used the assumption that the additive noise $v[n]$ is white (more specifically, uncorrelated with the signal $x[n]$).

Using the assumption that the initial phase ϕ_i of the signal $x[n]$ is i.i.d. and uniformly distributed, we get

$$\begin{aligned} E\left\{\alpha_i e^{j(\omega_i n+\phi_i)}\left(\alpha_l e^{j(\omega_l n+\phi_l)}\right)^*\right\} &= E\left\{\alpha_i e^{j(\omega_i n+\phi_i)}\alpha_l e^{-j(\omega_l n+\phi_l)}\right\} \\ &= E\left\{\alpha_i\alpha_l e^{j(\omega_i-\omega_l)n} e^{j\phi_i} e^{-j\phi_l}\right\} \\ &= \alpha_i\alpha_l e^{j(\omega_i-\omega_l)n} E\left\{e^{j\phi_i} e^{-j\phi_l}\right\} \\ &= \begin{cases} 0, & i\neq j \\ \alpha_i^2, & i=j \end{cases} \end{aligned}$$

where we have used the fact that

$$E\left\{e^{j\phi_i}\right\} = E\{\cos(\phi_i) + j\sin(\phi_i)\} = 0$$

Therefore

$$R_x = E\{\mathbf{x}[n]\mathbf{x}[n]^*\} = \begin{bmatrix} \alpha_1^2 & 0 & \ldots & 0 \\ 0 & \alpha_2^2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots \\ 0 & \ldots & \ddots & \alpha_K^2 \end{bmatrix} = diag\left(\alpha_1^2, \alpha_2^2, \ldots, \alpha_K^2\right)$$

Eq. (7) now simplifies to

$$R_y = AR_xA^* + \sigma^2 I \tag{8}$$

We see now $R_x \in \mathbb{R}^{K\times K}$ has K positive eigenvalues. $AR_xA^* \in \mathbb{R}^{M\times M}$ (note the increase of dimensions) therefore has K positive eigenvalues $\{\mu_i\}_{i=1}^K$ and $M-K$ zero eigenvalues. Thus

$$eig\left(R_y\right) \triangleq \lambda_i = \begin{cases} \mu_i + \sigma^2, & i = 1,2,\ldots,K \text{ (signal space)} \\ \sigma^2, & i = K+1,\ldots,M \text{ (noise space)} \end{cases} \tag{9}$$

Denoting the eigen pairs of R_y as $\{\mathbf{q_i}, \lambda_i\}_{i=1}^M$, we have the eigenvector matrix

$$\underbrace{Q}_{M\times M} = \left[\begin{array}{ccc|c} \mathbf{q_1} & \mathbf{q_2} & \ldots\mathbf{q_K} & \mathbf{q_{K+1}}\ldots\mathbf{q_M} \end{array}\right] \triangleq \left[\underbrace{Q_s}_{M\times K}, \underbrace{Q_n}_{M\times(M-K)}\right]$$

where Q_s denotes the signal eigenvector matrix and Q_n denotes the noise eigenvector matrix. *We have thus "separated" the signal space from the noise space! This is why this MUSIC algorithm belongs to the so-called subspace method.*

2.3 MUSIC: Algorithm

The fundamental fact in MUSIC: Using Eq. (9), we have

$$R_y\mathbf{q}_i = \sigma^2\mathbf{q}_i, \quad \forall i > K \tag{10}$$

Substituting in Eq. (8), we have

$$R_y\mathbf{q}_i = \left(AR_xA^* + \sigma^2 I\right)\mathbf{q}_i, \quad \forall i \tag{11}$$

Combining (10) and (11), we get

$$AR_xA^*\mathbf{q}_i = \mathbf{0}, \quad \forall i > K$$

Since $AR_x \in \mathbb{C}^{M\times K}$ is full column rank, its columns are thus linearly independent from each other. The above equation can then be simplified to

$$A^*\mathbf{q}_i = \mathbf{0}, \quad \forall i > K$$

Bring in now the definition of $A = \left[\begin{array}{cccc} \mathbf{a}\left(\omega_1\right) & \mathbf{a}\left(\omega_2\right) & \ldots & \mathbf{a}\left(\omega_K\right) \end{array}\right]$, we have

$$\mathbf{q}_i^*\mathbf{a}\left(\omega_k\right) = 0, \quad \forall\ K+1 \le i \le M;\ 1 \le k \le K$$

Recall that we are interested in finding the $\{\omega_k\}_{k=1}^K$. The MUSIC algorithm now simplifies to finding the ω such that

$$Q_n^*\mathbf{a}\left(\omega\right) = 0$$

In practice, the above is approximatly true due to the presence of errors in deriving Q_n. We instead use a more convenient method of finding the peaks in

$$P_{MUSIC}\left(\omega\right) = \frac{1}{\mathbf{a}\left(\omega\right)^* Q_nQ_n^*\mathbf{a}\left(\omega\right)} = \frac{1}{\left\|Q_n^*\mathbf{a}\left(\omega\right)\right\|_2}$$

MUSIC algorithm summary:

- step 1: take N samples of $y\left[n\right]$.

- step 2: compute the $M \times M$ sample covariance matrix estimation[1]

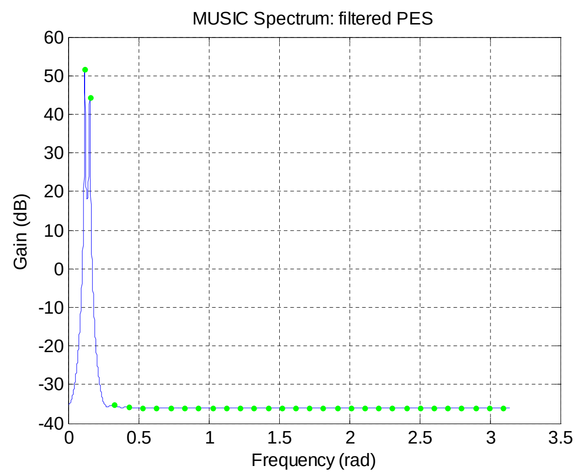
$$\hat{R}_y = \frac{1}{N-M+1} \sum_{n=M-1}^{N-1} y[n]\, y[n]^*$$

- step 3: form the eigen decomposition of $\hat{R}_y$, get the noise eigenvector space Q_n.
- step 4: estimate the number of narrow-band components K (if it is unknown), via looking the the eigenvalues of $\hat{R}_y$.
- step 5: find the K highest peaks of $P_{MUSIC}(\omega)$ by "sweeping" ω in the desired frequency range.

2.4 Simulation Result

2.4.1 Frequency Estimation

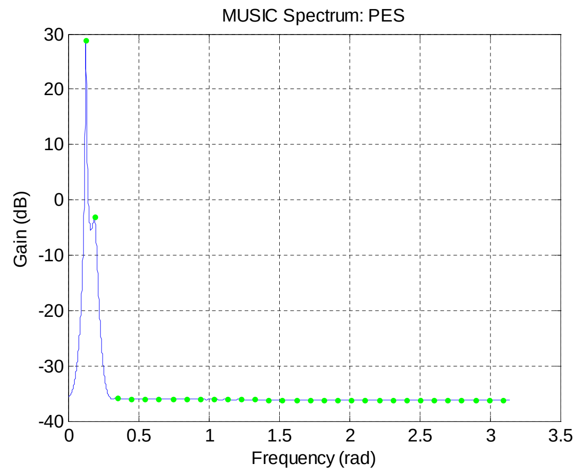
Narrow-band frequencies: 504 Hz and 648 Hz; signal length: 440 samples



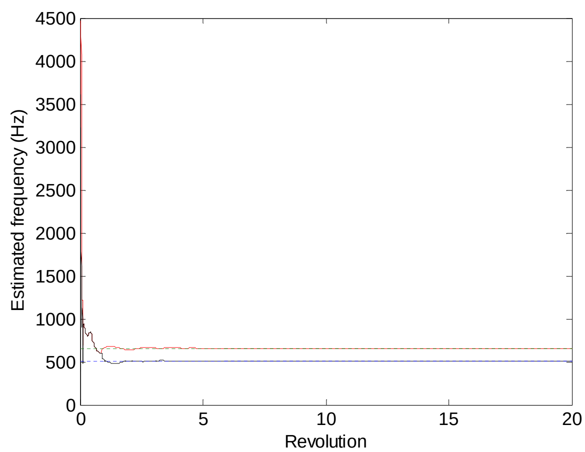
"clean" signal

Note: (1) I have applied a short script to locate the peaks in the above figure. (2) In practice, it is not necessary to sweep ω from 0 to π. In this example, we can pre-specify ω to be in the region [0, 0.3]. (using as much as possible the *a priori* information)

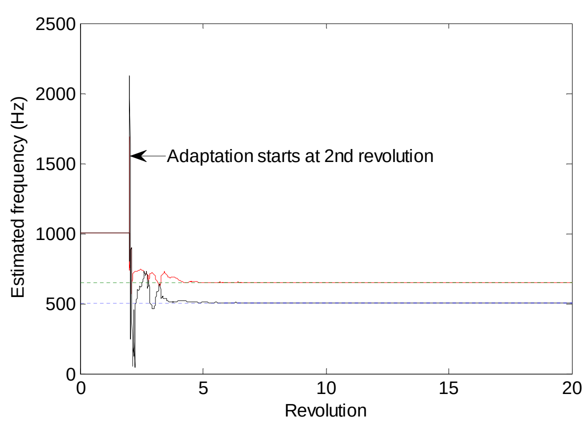
[1] M is a design parameter. $M > K$ is required to separate the signal and noise spaces. For a fixed N, a smaller M gives better estimated $\hat{R}_y$. However, for a given $\hat{R}_y$, a larger M gives the better separation of the signal and noise spaces. For systems high sampling frequency ω is usually small and very close to each other ($\omega = 2\pi T_s \Omega$, where Ω is in Hz.). $M = K^3$ is seen to give good performance in such systems.



signal in colored noise

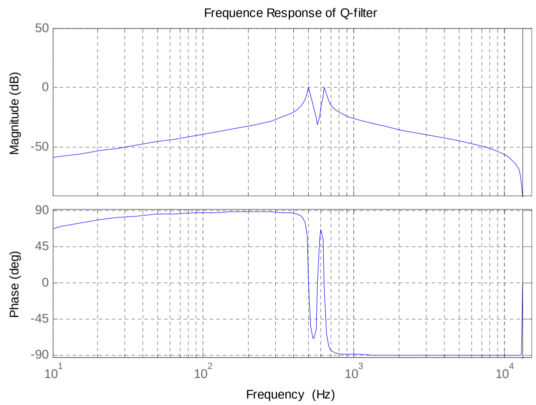


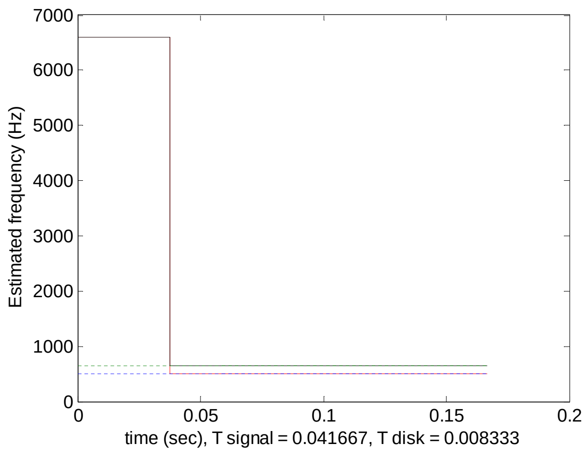
IMP method introduced in 2010 CML meeting

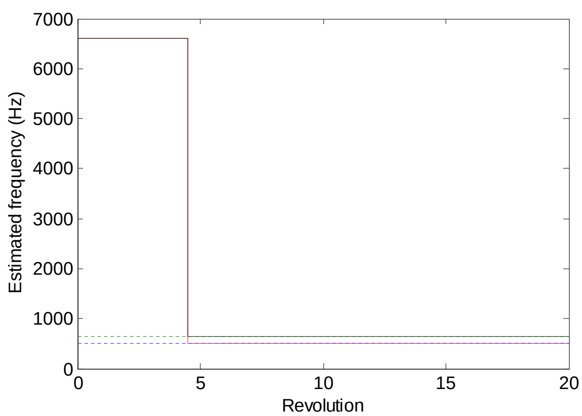


Direct DOB method

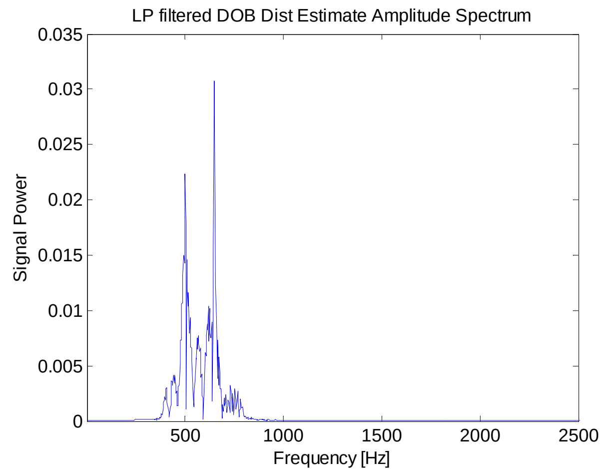
2.4.2 Narrow-band Disturbance Rejection

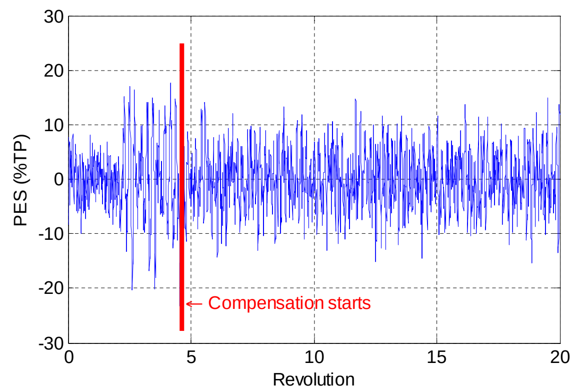






330 samples of data collected for MUSIC


LP filtered DOB Dist Estimate Amplitude Spectrum
Signal Power
0.035
0.03
0.025
0.02
0.015
0.01
0.005
0
500
1000
1500
2000
2500
Frequency [Hz]


PES (%TP)
30
20
10
0
-10
-20
-30
← Compensation starts
0
5
10
15
20
Revolution